

KIRKKONUMMI MAKILUOTO, Finland

WMO: 027940

Lat: 59.920N

Lon: 24.349E

Elev: 7

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.9	8.5	-0.8	5.3	3.6	4.6	7.0	10.6	41.3	35.0	39.0	34.1	16.1	30	0.854

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	75.1	68.8	72.4	67.1	70.0	65.3	70.1	74.0	68.2	71.9	65.9	69.3	12.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	104.7	73.1	66.7	98.1	71.2	64.3	90.1	68.6	34.1	74.2	32.5	72.0	30.7	69.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
37.7	33.9	30.7	2.2	77.6	8.1	4.5	-3.7	80.8	-8.4	83.4	-13.0	86.0	-18.9	89.3		
			1.7	70.6	8.4	2.8	-4.3	72.6	-9.3	74.2	-14.0	75.8	-20.1	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	26.6	26.9	30.7	38.2	48.4	56.6	64.0	63.0	56.1	45.7	39.2	33.8
	DBStd	14.28	8.17	9.13	5.97	3.84	5.56	4.01	4.88	4.09	4.55	5.38	6.23	7.15
	HDD50	3411	725	647	599	353	97	3	0	0	10	149	325	503
	HDD65	7664	1190	1067	1064	803	514	251	79	87	268	598	775	968
	CDD50	1296	0	0	0	0	48	202	435	404	191	16	0	0
	CDD65	74	0	0	0	0	0	0	48	26	0	0	0	0
	CDH74	125	0	0	0	0	0	2	98	25	0	0	0	0
	CDH80	4	0	0	0	0	0	0	3	1	0	0	0	0
Wind	WSAvg	16.3	18.0	16.9	16.6	14.2	13.8	14.4	13.1	14.2	15.9	18.2	19.6	20.9
Precipitation	PrecAvg	22.8	1.8	1.3	1.1	1.1	1.2	1.9	2.3	2.9	2.0	2.6	2.5	1.9
	PrecMax	30.1	2.9	3.7	2.1	2.3	2.9	3.4	5.6	6.0	3.5	4.3	6.1	3.3
	PrecMin	17.3	0.2	0.0	0.2	0.2	0.1	0.6	0.2	0.3	0.7	1.3	0.4	0.5
	PrecStd	2.9	0.7	0.9	0.5	0.6	0.9	0.7	1.4	1.3	0.9	0.8	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	39.2	43.1	51.8	66.5	69.8	80.0	76.7	66.9	57.4	50.1	46.3
		MCWB	39.1	36.5	37.8	45.1	56.1	61.1	71.0	69.8	61.5	53.9	48.7	45.1
	2%	DB	39.5	37.7	40.2	48.2	62.5	67.0	76.6	73.7	64.9	55.4	48.7	44.8
		MCWB	38.3	36.2	36.9	43.2	55.1	60.3	69.5	68.0	60.4	53.1	47.5	43.8
	5%	DB	38.1	36.6	38.4	45.8	59.2	64.8	74.1	71.2	63.4	54.1	47.4	43.1
		MCWB	37.0	35.5	35.9	41.5	53.3	59.5	68.3	66.0	59.6	52.0	46.2	41.9
	10%	DB	36.2	35.6	36.9	43.7	56.4	62.9	71.8	69.1	61.9	52.8	46.3	41.7
		MCWB	35.2	34.7	35.1	40.2	51.5	58.1	67.0	64.4	58.9	50.7	44.9	40.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	37.7	39.2	45.7	58.1	63.9	72.7	71.3	63.0	55.3	49.2	45.6
		MCDB	40.1	38.4	42.0	50.7	64.5	67.2	77.5	75.2	65.4	56.6	49.7	45.9
	2%	WB	38.5	36.8	37.6	43.8	56.2	62.0	71.0	69.4	61.6	53.9	47.9	43.9
		MCDB	39.3	37.3	39.3	47.5	61.3	65.7	74.9	72.9	63.6	54.9	48.6	44.3
	5%	WB	37.2	35.7	36.6	42.2	54.2	60.4	69.3	67.2	60.4	52.6	46.7	42.1
		MCDB	38.1	36.3	38.0	45.2	58.6	63.9	73.3	70.6	62.4	53.7	47.3	42.9
	10%	WB	35.6	34.8	35.5	40.7	52.2	58.8	67.7	65.1	59.4	51.1	45.2	40.6
		MCDB	36.4	35.5	36.7	43.2	55.9	62.3	71.5	68.3	61.5	52.5	46.1	41.7
Mean Daily Temperature Range	5% DB	MDBR	6.9	6.1	7.1	7.1	8.7	8.9	9.3	8.7	7.6	6.6	5.0	5.7
		MCDBR	5.1	4.4	7.1	10.0	12.3	11.3	11.4	10.3	9.1	6.9	5.3	5.4
		MCWBR	5.1	4.1	5.0	6.3	6.9	6.7	6.4	6.5	6.1	5.6	5.5	5.5
	5% WB	MCDBR	5.2	4.3	6.3	9.0	11.1	10.1	9.8	8.7	7.8	6.1	5.1	5.2
		MCWBR	5.3	4.1	4.8	5.9	7.1	6.6	5.9	6.0	5.8	5.8	5.5	5.6
Clear-Sky Solar Irradiance	taub	0.279	0.312	0.334	0.365	0.362	0.359	0.384	0.378	0.349	0.335	0.309	0.264	
	taud	2.142	2.194	2.255	2.317	2.402	2.474	2.429	2.444	2.511	2.400	2.221	2.087	
	Ebn at Noon	153	214	251	264	275	277	266	259	248	209	141	110	
	Edh at Noon	14	23	30	34	33	31	32	30	23	19	13	10	
All-Sky Solar Radiation	RadAvg	79	257	733	1228	1698	1770	1697	1324	806	350	96	48	
	RadStd	13	46	93	130	181	132	151	133	103	52	17	7	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (57 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-241	N/A	N/A	N/A

Nomenclature: See separate page